Horses flexibly adjust their multimodal communication across social contexts

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Socially complex primate species have correspondingly complex and flexible communication systems, combining multiple modalities and showing context-dependent variation in signal complexity. This study examines multimodal communication in horses, a species that also lives in complex multilevel societies. Based on primate research, we predicted greater communicative complexity, measured as the number of signals and modalities used, in contexts involving higher social uncertainty (e.g., play, aggression, greeting). Overall, the number of signals and modalities varied according to social context. In line with our hypotheses, horses produced more signals in higher-uncertainty contexts and also during grooming. In contrast, the degree of multimodality varied weakly with uncertainty, although it peaked in greeting, *a priori* classified as highly uncertain. Signal number varied with the level of social uncertainty, whereas the degree of multimodality was higher in close-range affiliative and investigative contexts. Identifying such flexibility in a non-primate species provides a comparative basis for investigating the evolutionary drivers of complex communication.

1. Introduction

Human communication shows remarkable complexity: we combine signals of different types and modalities (for instance, speech with gesture and facial expressions) to produce diverse multimodal combinations across social situations (Fröhlich et al., 2025). Human communication is also highly flexible: we modify our communicative complexity through both the number and type of signals according to the context and our interlocutors. For example, speakers adjust the number of facial and gestural cues depending on environmental noise and the degree of shared knowledge between partners (Rasenberg et al., 2022). They also produce fewer gestures when addressing a socially distant superior than when speaking to a close friend, despite conveying the same content (Brown et al., 2023). This flexibility in combining and adjusting signals across modalities

appears central to human communication and offers a useful point of comparison with other species to better understand its evolutionary roots.

Non-human primates likewise display flexibility in their use of signals of different types and sensory modalities. For example, captive mangabeys flexibly modulate the types of signals and modalities used in their communicative sequences according to the social context and the characteristics of the signaller, such as age and sex (Aychet et al., 2021). Similarly, chimpanzees increase their communicative complexity in contexts of heightened social uncertainty, such as post-conflict approaches, long-distance communication, and subgroups' fusion. In these situations, they produce a greater number of signals across sensory modalities ("multisensory signals") and more signals overall, either simultaneously or sequentially (so-called "combined signals"), likely reflecting an increased need to clarify intentions when interaction outcomes are more unpredictable (Grampp et al., 2023).

Although communicative flexibility has been documented beyond primates, research has primarily focused on a limited set of other taxa, including birds and cetaceans (Cusano, 2021; Fröhlich et al., 2025), whereas ungulates remain largely understudied (but see Soltis et al., 2014 on elephants). This bias limits our understanding of its distribution across taxa and its evolutionary origins. To address this, we must examine species with distinct evolutionary trajectories. Identifying whether such flexibility extends beyond these taxa is directly relevant to understanding the broader evolutionary foundations within which human language evolved.

Horses offer an ideal model for this comparative approach. They rely on multiple communicative modalities, including visual, tactile, olfactory and auditory signals (Hothersall et al., 2010; Krueger et al., 2011; Lemasson et al., 2009; McDonnell, 2003; Proops et al., 2009; Shimada & Suzuki, 2020; Wathan et al., 2015). Additionally, their wild counterparts live in socially complex groups that exhibit fission–fusion dynamics (Maeda et al., 2021; Ozogány et al., 2023). Living in social groups provides both short-term benefits (e.g., protection from predators; Silk, 2007) and long-term advantages (e.g., improved offspring survival; Silk, 1979). However, group living also presents challenges: individuals must coordinate interactions and convey information about identity, emotional state, and motivation (Freeberg et al., 2012; Peckre et al., 2019). This has led to the social complexity hypothesis for communicative complexity (SCHCC; Freeberg et al., 2012), which proposes that species living in more socially complex groups evolve more elaborate communication systems to navigate these challenges. The SCHCC has been posited as a potential driver for the evolution of human language (Freeberg et al., 2012).

Recent work further suggests that the SCHCC also applies to situations of social uncertainty, where unpredictable or ambiguous interactions increase the need for communicative precision (Grampp et al., 2023). The SCHCC has been widely tested in non-human primates with varying social complexity levels (Bouchet et al., 2013; Clark et al., 2022; Grampp et al., 2023; Rebout et al., 2020; Rincon et al., 2023).

Based on the SCHCC, it is therefore predicted that horses possess a rich and flexible communicative repertoire. Accordingly, they may vary the number of signals and modalities they use depending on the social context. We predicted greater communicative complexity—reflected in an increased number of signals and modalities—in contexts involving higher social uncertainty or risk of conflict or escalation, such as play and aggression, compared with lower-risk contexts like affiliative contact or follow (Table 1).

2. Methodology

2.1. Data collection

We analysed a video dataset of naturally occurring social interactions among 56 domestic horses of mixed breeds, sexes, and ages (2–23 years), filmed across several sites in southern England between 2020 and 2025. Recordings were obtained through 30-minute focal samples and ad libitum observations. The final dataset comprised 328 social interactions (mean ± SD duration per interaction: 7.95 ± 10.17 s).

2.2 Video coding

Each interaction was coded frame by frame in ELAN 7.0, with every simultaneously co-occurring visual, acoustic, tactile, and olfactory signal annotated using an ethogram of a total of 130 possible actions (e.g., blink, nose-to-nose, tail lifted) derived from published equid ethograms (Hex & Rubenstein, 2024; McDonnell, 2003) and the EquiFACS (Equine Facial Action Coding System) manual (Wathan et al., 2015). Auditory signals were annotated when they consisted of clearly audible and identifiable vocalisations or sounds produced by the individual (e.g., tail swishing). Tactile signals were annotated when clear physical contact occurred between individuals. Olfactory behaviours were annotated only when their production was unambiguous, such as during direct sniffing of another individual (e.g., nose-to-nose investigation with clear nasal dilation) or during defecation and urination.

The social context and the identity of both signaller and receiver were also coded. When possible, interactions were coded twice so that each individual could be coded as both signaller and receiver. Inter-rater reliability (Cohen's $\kappa = 0.86$)

confirmed consistent signal identification for full-body behaviours, and facial behaviours were coded by an EquiFACS-certified coder.

The coded data was segmented into time slices of discrete signal combinations (e.g., left ear forward + eyes open). A new signal combination was recorded every time at least one action changed (e.g., moving the left ear from a forward position to a backward position). Individuals often modified their signalling over the course of a single social interaction, producing multiple signal combinations. Given that the ethogram comprised 130 possible actions, the number of discrete signal combinations was considerable, and the final dataset comprised 15,118 unique signal combinations.

2.3 Statistical analyses

Communicative complexity was quantified across twelve social contexts (Table 1) using two indicators of complexity: the number of signals per signal combination and the number of modalities per signal combination. To test predictions derived from the SCHCC, contexts were classified *a priori* into higher and lower social uncertainty categories, based on risk of conflict or escalation. Agonistic interactions carry a clear risk of physical injury and negative consequences for social relationships (Aureli et al., 2002), whereas play requires signals that prevent misinterpretation as aggression, which could otherwise lead to escalation (Bekoff & Byers, 1998; Palagi et al., 2015). In primates, both of these contexts have been described as requiring clearer signalling compared to other social contexts to reduce costs of ambiguity (e.g., through increased use of multimodal combinations; Doherty et al., 2023). Greeting was also classified as highly uncertain based on evidence from fission-fusion primates that greetings might be involved in tension reduction upon reunion and in the reassessment of social and dominance relationships (Fedurek et al., 2021). In plains zebras, overlap between greeting and agonistic signals suggests potential ambiguity and an elevated risk of escalation in this context (Hex & Rubenstein, 2025).

Table 1. Classification of social contexts into higher and lower social uncertainty categories used in the statistical analyses.

Higher social uncertainty	*Lower social uncertainty*
Chase/Charging (high-intensity, directed pursuit without physical contact)	Neutral approach (approach without overt threat signals or displacement)
Fight (high-intensity aggression involving physical contact)	Follow (sustained movement maintaining proximity behind another individual)
Threat (overt agonistic	Olfactory investigation (nose-

display)	to-nose sniffing during ongoing proximity)
Displace (approach resulting in the other horse moving away)	Affiliative contact
Withdraw (often occurring in response to a threat)	Grooming
Play	
Greeting (nose-to-nose sniffing after being separated for more than five minutes)	

Greeting and olfactory investigation were distinguished by whether nose-to-nose sniffing occurred following a period of separation between partners, with greeting classified as higher uncertainty and olfactory investigation as lower uncertainty since the individuals involved were already in close proximity.

To analyse variation in signal number, we fitted generalised linear mixed models (GLMMs) with a Poisson distribution (log link) using the lme4 package (v1.1.37) in R (v4.4.3; R Core Team). Signal number was modelled as a function of social context (categorical predictor), with interaction ID and signaller–receiver pair ID included as random effects. To directly test predictions derived from the SCHCC, a complementary model was fitted in which social contexts were grouped by social uncertainty (higher vs. lower), and the uncertainty category was used as the predictor, while retaining the same random-effects structure.

Variation in the number of modalities per signal combination was analysed using cumulative link mixed models (CLMMs) with a logit link, implemented in the ordinal package (v2023.12.4.1). Modality count was treated as an ordinal response variable. As above, models tested both social context and social uncertainty as predictors. Due to convergence constraints, the random-effects structure for these models was simplified to include interaction ID only.

3. Results

3.1. Signal number across contexts and uncertainty levels

The full model was a significantly better fit than the null model (Poisson GLMM; LRT: $\chi^2(11) = 69.39$, $p < 0.001$), with social context significantly predicting the number of signals per combination (Wald $\chi^2(11) = 79.10$, $p < 0.001$; Fig. 1A). When contexts were grouped by social uncertainty, the model also outperformed the null model (Poisson GLMM; LRT: $\chi^2(1) = 33.67$, $p < 0.001$), with uncertainty category significantly predicting signal number (Wald $\chi^2(1) = 35.99$, $p < 0.001$), and highly uncertain contexts associated with a greater number of signals (Fig.

1B).

(A) (B)

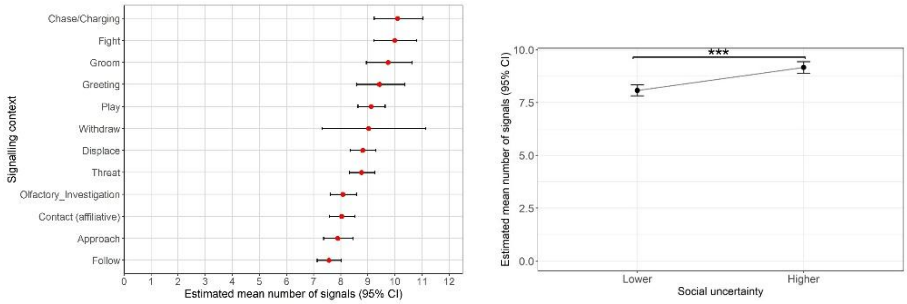

Figure 1. Signal number per signal combination across social contexts and social uncertainty. (A) Variation across the twelve social contexts studied. (B) Comparison between lower and higher social uncertainty contexts. *** $p < 0.001$ Points show model-estimated marginal means from Poisson generalized linear mixed models and error bars indicate 95% confidence intervals.

3.2. Degree of multimodality across contexts and uncertainty levels

The full model was a significantly better fit than the null model (cumulative link mixed model; $\chi^2(11) = 171.84$, $p < 0.001$), indicating that social context predicted the number of modalities per signal combination (Fig 2A). When contexts were grouped by social uncertainty, the model also outperformed the null model (cumulative link mixed model; LRT: $\chi^2(1) = 9.45$, $p = 0.002$), with highly uncertain contexts associated with a lower degree of multimodality ($\beta = -1.31 \pm 0.42$ SE, $z = -3.09$, $p = 0.002$; Fig. 2B), although the magnitude of this difference was small ($\Delta \approx 0.035$ modalities).

(A) (B)

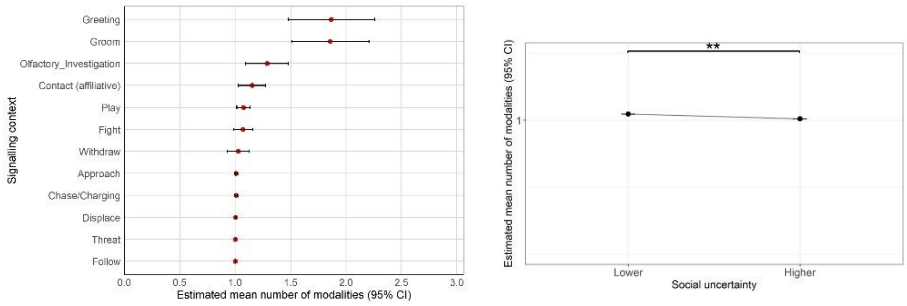

Figure 2. Degree of multimodality per signal combination across social contexts and social uncertainty levels. (A) Variation across the twelve social contexts. (B) Comparison between lower and higher social uncertainty contexts. ** $p < 0.01$

Points show model-estimated marginal means from cumulative link mixed models and error bars indicate 95% confidence intervals.

4. Discussion & Conclusion

Our analyses provide partial support for the predictions derived from the SCHCC. Horses adjusted their communicative complexity across social contexts, with the number of signals per signal combination increasing in contexts characterised by higher social uncertainty compared to lower-uncertainty contexts. Unexpectedly, grooming elicited the highest number of signals. This may be because, although grooming is typically considered a low-risk affiliative behaviour linked to bond maintenance and hygiene in horses (Hodgson et al., 2024), studies in feral populations show that grooming rates also rise in unstable groups where aggression is more frequent (Sigurjónsdóttir & Haraldsson, 2019). It is therefore possible that grooming may additionally serve a tension-reducing function, occurring more often when social uncertainty or the risk of aggression within a group is higher.

In contrast to signal number, the number of modalities used differed only marginally between uncertainty categories, remaining close to unimodal signalling despite statistical significance. Inspection of context-level estimates indicated that the degree of multimodality was highest in greeting and grooming, as well as in other close-range affiliative or investigative contexts. In greeting, increased multimodality may reflect greater demands for information exchange following periods of separation, whereas in grooming it may relate to the need for tactile contact, fine-scale coordination and mutual adjustment between partners (e.g., opposite head orientations and matching of groomed body areas; Fröhlich, 2017).

Overall, these findings indicate that horses possess a multimodal communicative system in which communicative complexity varies across contexts in distinct ways. Variation in signal number appears linked to social uncertainty, whereas the degree of multimodality may be shaped by other social demands. Thus, horses exhibit communicative flexibility, with at least one aspect of their communication varying in line with predictions of the SCHCC.

Acknowledgments

The first author was supported by a doctoral training grant (ESRC; grant reference ES/P000673/1) from the South Coast Doctoral Training Partnership. We are grateful to the Sparsholt College Equine Centre, the Donkey Sanctuary, and the Hill Livery and Three Greys horse riding centres for allowing access to their animals and facilities.

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